LECTURE 06

Equivalence Relations

Equivalence classes, modular arithmetic and partitions

Definition

> An **equivalence relation** on a set S is a relation on S which is **reflexive**, **symmetric** and **transitive**.

Examples

Parity. Let $S = \mathbb{Z}$ and define $R = \{(x, y) \mid x \text{ and } y \text{ have the same parity}\}$, i.e. x and y are either both even or both odd. The parity relation is an equivalence relation.

1. For any $x \in \mathbb{Z}$, x has the same parity as itself, so $(x, x) \in R$.
2. If $(x, y) \in R$, then x and y have the same parity, so $(y, x) \in R$.
3. If $(x, y) \in R$ and $(y, z) \in R$, then x and z have the same parity as y, so they have the same parity as each other (if y is odd, both x and z are odd; if y is even, both x and z are even), thus $(x, z) \in R$.

The square relation. Let $S = \mathbb{R}$ and define $R = \{(x, y) \mid x^2 = y^2\}$. The square relation is an equivalence relation.

1. For all $x \in \mathbb{R}$, $x^2 = x^2$, so $(x, x) \in R$.
2. If $(x, y) \in R$, then $x^2 = y^2$, so $y^2 = x^2$ and $(y, x) \in R$.
3. If $(x, y) \in R$ and $(y, z) \in R$, then $x^2 = y^2 = z^2$, so $(x, z) \in R$.

The identity relation. For any set S, the identity relation on S is $I_S = \{(x, x) \mid x \in S\}$. This is an equivalence relation for rather trivial reasons.

1. Obvious.
2. If $(x, y) \in R$ then $y = x$, so $(y, x) = (x, x) \in R$.
3. If $(x, y) \in R$ and $(y, z) \in R$ then $x = y = z$, so $(x, z) = (x, x) \in R$.

Modular Arithmetic

Let $S = \mathbb{Z}$. For each positive integer m, we define the **modular relation** $\underset{m}{\equiv}$ by $x \underset{m}{\equiv} y$ iff $m \mid (x - y)$, i.e.

$$\underset{m}{\equiv} = \{(x, y) : m \mid x - y\}.$$

Examples.

$$\begin{array}{llll} 7 \underset{5}{\equiv} 2 & 11 \underset{5}{\equiv} 1 & 10 \underset{5}{\equiv} 0 & -12 \underset{5}{\equiv} 3 \\ 7 \underset{3}{\equiv} 1 & 11 \underset{3}{\equiv} 2 & 10 \underset{3}{\equiv} 1 & -12 \underset{3}{\equiv} 0 \end{array}$$

Another way to think about the modular relation is:

> $x \underset{m}{\equiv} y$ iff x and y have the same remainder when divided by m.

By the division algorithm, $x = mq_1 + r_1$ and $y = mq_2 + r_2$, so $x - y = m(q_1 - q_2) + (r_1 - r_2)$. Hence $m \mid x - y$ iff $m \mid r_1 - r_2$. Since $|r_1 - r_2| < m$, we get $m \mid r_1 - r_2$ iff $r_1 - r_2 = 0$ iff $r_1 = r_2$.

Theorem. For any natural number m, the modular relation $\underset{m}{\equiv}$ is an equivalence relation on $\mathbb{Z}$.

Proof. For any $x \in \mathbb{Z}$, since $x - x = 0$ and $m \mid 0$, we have $x \underset{m}{\equiv} x$. (Reflexivity)

If $x \underset{m}{\equiv} y$ then $m \mid x - y$. Since $y - x = -(x - y)$, we get $m \mid y - x$, and so $y \underset{m}{\equiv} x$. (Symmetry)

If $x \underset{m}{\equiv} y$ and $y \underset{m}{\equiv} z$ then $m \mid x - y$ and $m \mid y - z$. Since

$$x - z = (x - y) + (y - z),$$

we have $m \mid x - z$, so $x \underset{m}{\equiv} z$. (Transitivity)

Equivalence Classes

Given an equivalence relation R on a set S, we define the **equivalence class containing an element** x of S by

$$[x]_R = \{y \mid (x, y) \in R\} = \{y \mid xRy\}.$$

The text uses the notation x/R (which I am not fond of) for what I have called $[x]_R$.

Examples

- Let $S = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Then $[1] = \{1, 2\}$, $[2] = \{1, 2\}$, $[3] = \{3\}$.
- Let $S = \mathbb{R}$ and $R = \{(x, y) \mid x^2 = y^2\}$. Then $[0] = \{0\}$, $[1] = \{1, -1\}$, $\left[\frac{1}{4}\right] = \left\{\frac{1}{4}, -\frac{1}{4}\right\}$, $[x] = \{x, -x\}$.

More Examples

- Let $S = \mathbb{Z}$ and $R = \{(x, y) \mid x \text{ and } y \text{ have the same parity}\}$.

$$[0] = [2] = \cdots = [2k] = \{0, \pm 2, \pm 4, \pm 6, ..., \pm 2k, ...\}$$
$$[-1] = [1] = \cdots = [2k + 1] = \{\pm 1, \pm 3, \pm 5, ..., \pm 2k + 1, ...\}$$

- For any set S, $I_S = \{(x, x) \mid x \in S\}$. Then $[a] = \{a\}$ for all $a \in S$.
- Let $S = \mathbb{Z}$ and $R =$ "$\underset{5}{\equiv}$".

$$[0] = \{0, \pm 5, \pm 10, \pm 15, ..., 5k\} \quad (k \in \mathbb{Z})$$
$$[1] = \{..., -9, -4, 1, 6, 11, ..., 5k + 1\}$$
$$[2] = \{..., -8, -3, 2, 7, 12, ..., 5k + 2\}$$
$$[3] = \{..., -7, -2, 3, 8, 13, ..., 5k + 3\}$$
$$[4] = \{..., -6, -1, 4, 9, 14, ..., 5k + 4\}$$

Properties of Equivalence Classes

Let R be an equivalence relation on the set S.

I. For all $x \in S$, $x \in [x]$.

Since R is reflexive, $(x, x) \in R$ for all $x \in S$.

II. If $y \in [x]$ then $x \in [y]$, and $[x] = [y]$.

Since R is symmetric, if $y \in [x]$ then $(x, y) \in R$ so $(y, x) \in R$ and we have $x \in [y]$. If $s \in [x]$, then $(x, s) \in R$, so $(s, x) \in R$ and then $(s, y) \in R$ (by transitivity) and finally $(y, s) \in R$, so $s \in [y]$. Similarly, if $t \in [y]$ then $t \in [x]$ and so $[x] = [y]$.

III. For any x and $y \in S$, either $[x] = [y]$ or $[x] \cap [y] = \varnothing$.

If there is a $z \in [x]$ which is also in $[y]$, then $(x, z) \in R$ and $(y, z) \in R$. By symmetry, $(z, y) \in R$ as well. By transitivity, $(x, y) \in R$, so $y \in [x]$. By II, $[x] = [y]$.

An Important Equivalence Relation

Let S be the set of fractions:

$$S = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Define a relation R on S by

$$\frac{a}{b} \, R \, \frac{c}{d} \quad \text{iff} \quad ad = bc.$$

This relation is an equivalence relation.

1. For any fraction a/b, $a/b \, R \, a/b$ since $ab = ba$. (Reflexivity)
2. If $a/b \, R \, c/d$, then $ad = bc$, so $cb = da$ and $c/d \, R \, a/b$. (Symmetry)
3. If $a/b \, R \, c/d$ and $c/d \, R \, e/f$, then $ad = bc$ and $cf = de$. Multiply the first equation by f to get $adf = bcf$, so $adf = bde$. Divide by d (which is not 0) to get $af = be$, so $a/b \, R \, e/f$. (Transitivity)

The equivalence classes of this equivalence relation, for example

$$\left[\frac{1}{1}\right] = \left\{ \frac{2}{2}, \frac{3}{3}, \ldots, \frac{k}{k}, \ldots \right\}$$
$$\left[\frac{1}{2}\right] = \left\{ \frac{2}{4}, \frac{3}{6}, \frac{4}{8}, \ldots, \frac{k}{2k}, \ldots \right\}$$
$$\left[\frac{4}{5}\right] = \left\{ \frac{4}{5}, \frac{8}{10}, \frac{12}{15}, \ldots, \frac{4k}{5k}, \ldots \right\}$$

are called **rational numbers**. The set of all the equivalence classes is denoted by $\mathbb{Q}$.

Partitions

A **partition** of a set S is a family F of **non-empty** subsets of S such that

(i) if A and B are in F then either $A = B$ or $A \cap B = \varnothing$, and
(ii) $\bigcup_{A \in F} A = S$.

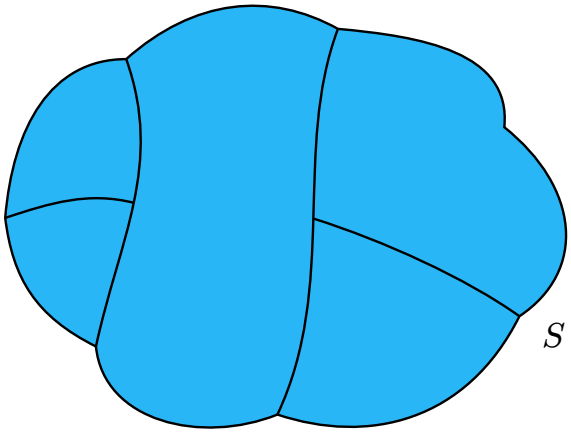


Figure 1: A partition of S into pairwise disjoint, non-empty blocks whose union is S.